

**National Institute of Technology Kurukshetra**  
**[B.Tech. (EE-Branch) II- Semester**  
**Mid-Semester-Exam-1**

**[Advanced Engineering Physics(AEP); Course No. PHIC104]**

**Note: All questions are compulsory**

Time allotted: 50 min.

Max. Marks: 15

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| 1. | <p>(a) For <math>A = (ax-y)\hat{i} + (bx+z)\hat{j} + (cy-x)\hat{k}</math>, find a, b, c such that field is irrotational and solenoidal.</p> <p>(b) Why is the electric field zero inside a hollow charged conductor? Explain how this principle is used in Faraday cages, aircraft safety, and EMI shielding.</p> <p>(c) Why is a dielectric preferred over a conductor as an insulating material in electrical equipment, even though both respond to electric fields?</p> <p style="text-align: center;">OR</p> <p>Explain polarization of a dielectric placed in an external electric field. Explain in terms of displacement vector.</p> | <p>[3]</p> <p>[1,1]</p> <p>[2.5]</p>  |
| 2. | <p>(a) State the necessary condition for a material to become magnetic. Explain with examples.</p> <p>(b) Why is magnetic field intensity preferred over magnetic flux density while studying magnetic properties of materials? The magnetization of a specimen is 600 A/m when applied field is 300 A/m. Find the susceptibility and identify the material.</p> <p>(c) Why diamagnetic materials are used in NMR and MRI?</p>                                                                                                                                                                                                               | <p>[2.5]</p> <p>[2, 1]</p> <p>[2]</p> |